

1. What is a function? State its types and explain the syntax to define a function.

Answer:

A **function** is a block of reusable code that performs a specific task. It helps to reduce code repetition and makes the program easier to understand and maintain.

Types of functions in Python:

1. **Built-in Functions** – Functions already provided by Python, such as `print()`, `len()`, and `type()`.
2. **User-defined Functions** – Functions created by the programmer according to their requirements.

Syntax:

```
def function_name(parameters):  
    # Function body  
    statements
```

Example:

```
def add(a, b):  
    result = a + b  
    return result  
  
print(add(10, 20))
```

Here, `def` is used to define a function, `add` is the function name, and `a`, `b` are parameters.

2. What is a lambda function?

Answer:

A **lambda function** is a small, anonymous function in Python that is defined using the `lambda` keyword. It can have any number of arguments but contains only **one expression**.

Syntax:

```
lambda arguments: expression
```

Example:

```
square = lambda x: x * x
```

```
print(square(5))
```

Output:

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Lambda functions are mainly used when a simple function is needed for a short period of time.